

Beyond Speed Rankings: Application-Layer Broadband Performance in Nine Southeast Asian Countries

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ABSTRACT

Southeast Asian broadband is usually judged by headline speed-index rankings, which say little about whether real applications work. Thailand ranked 12th worldwide for fixed broadband in June 2026, yet a 2023 national consumer survey found 81% of 2,924 respondents reporting connection problems [7]. We analyze 990 million speed tests — 229.4M from Ookla Open Data and 760.6M from M-Lab NDT7, Q1 2023 to Q4 2025 — across nine countries, and score each country-quarter’s median download speed against the bandwidth requirements of voice, HD video, UHD video and cloud gaming (bandwidth only; we do not assess latency, and both datasets are crowdsourced and user-initiated rather than a random sample). Our preliminary results show that median speeds clear the voice and HD-video bandwidth bars in every country, every quarter of 2024–2025. That uniformity collapses at higher tiers. Fixed broadband stratifies sharply by geography, with a 16× spread between the strongest and weakest capital regions, and against the cloud-gaming bandwidth bar the median reaches 100% in Singapore but only a peak of 37% in Indonesia and 31% in Myanmar. Work is ongoing on per-province modeling and on peak-hour behavior.

KEYWORDS

Internet measurement, broadband performance, Southeast Asia, Ookla, M-Lab NDT7, application requirements

1 MOTIVATION

National broadband quality in Southeast Asia is usually argued from two sources that disagree with each other: index rankings and user complaints. Thailand was ranked 12th worldwide for fixed broadband in Ookla’s Speedtest Global Index in June 2026 [5], while a 2023 survey by the Thailand Consumer Council found that 81% of 2,924 respondents had encountered connection problems, mostly slow or dropping connections [7]. A national mean download speed cannot reconcile those two observations, because it describes the network rather than the service a user is trying to run over it.

We therefore ask a different question: *how do Southeast Asian networks perform against the demands of the applications people actually use?* Rather than ranking countries by speed, we score every country-quarter against published bandwidth requirements and report how close each country’s median connection comes to clearing each bar. The mismatch above is not unique to Thailand, so we run the analysis across nine countries.

2 DATA AND METHOD

We use two complementary open datasets covering Thailand, Vietnam, Laos, Myanmar, Cambodia, Indonesia, the Philippines, Malaysia and Singapore, over twelve quarters (Q1 2023–Q4 2025):

- **Ookla Open Data** [4] — 229,367,221 tests, published as quarterly tiles of roughly 600 m×600 m, fixed and mobile. Ookla servers sit inside access networks and use parallel streams, so these tests characterize last-mile capacity.
- **M-Lab NDT7** [3] — 760,607,387 single-stream tests to off-net servers, closer to what a single streaming or browsing flow observes.

Tiles are assigned to first-level administrative provinces by point-in-polygon against geoBoundaries ADM1 [6]; NDT7 points are joined to the same polygons. Province averages are test-weighted, and we keep only province-quarters with at least 100 tests and at least 5 tiles. Tests with non-positive throughput or duration are dropped, and IP addresses are checked against ip-api.com to exclude hosting/proxy traffic. Because user-initiated speed tests are a biased sample [1], we report distributions and success rates rather than treating any single figure as a population mean. Coverage is uneven: Myanmar is missing 10.7% of fixed and 22% of mobile province-quarters, and some Vietnamese and Cambodian provinces lack mobile data in low-volume quarters — gaps we flag rather than interpolate.

We evaluate four application classes against the thresholds of Lübben and Misfeld [2]: voice (64 kbps), HD video streaming (5 Mbps), UHD video streaming (25 Mbps) and cloud gaming (44 Mbps). **Success rate** is a country’s quarterly median download speed as a percentage of the application’s *download-speed* threshold, capped at 100%, reported for the eight quarters Q1 2024–Q4 2025. We deliberately do not evaluate the 25 ms latency component of the cloud-gaming requirement: for six of our nine countries most M-Lab tests terminate abroad (Thai tests mainly in Singapore and Hong Kong), and only Singapore, Indonesia and the Philippines are served largely in-country, so measured round-trip time mostly reflects international transit rather than domestic access.

3 PRELIMINARY RESULTS

3.1 Median speeds clear the low-bandwidth bars

The median download speed clears the voice and HD-video bandwidth thresholds (a 100% score) in every country, in every quarter of 2024–2025. This holds for the lowest-tier markets in the set as firmly as for Singapore. Whatever the regional digital divide consists of, a national median failing to reach 5 Mbps is not part of it — though this says nothing about the tail of individual tests, or about latency, which we do not evaluate.

3.2 Fixed broadband stratifies by geography

Most capital regions outperform their own country (Figure 1): Bangkok reaches 298.8 Mbps against a national 276.9, Kuala Lumpur 212.3 against 194.8, and Manila NCR 159.6 against 132.0. Singapore

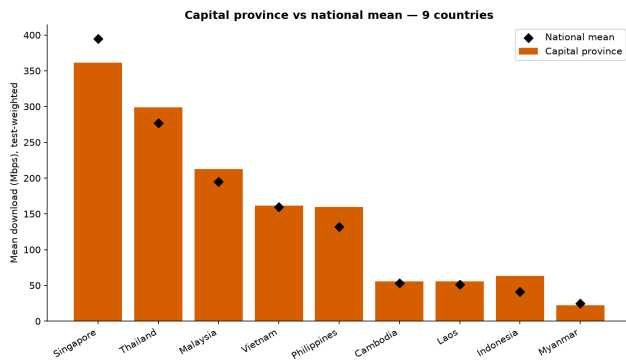


Figure 1: Capital province versus national mean download speed, fixed broadband, test-weighted, nine countries.

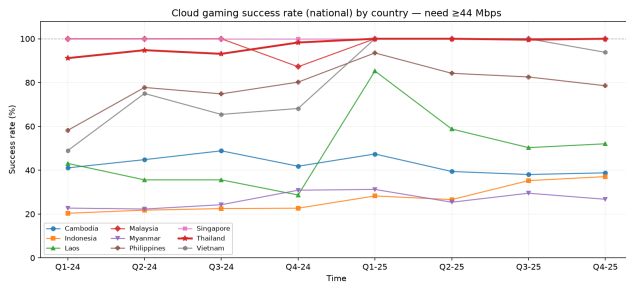


Figure 2: Cloud-gaming success rate (median download against the 44 Mbps threshold, capped at 100%) by country and quarter, Q1 2024–Q4 2025.

and Myanmar are the exceptions: Singapore’s Central Region (361.0) sits *below* its national mean (394.8), as does Yangon (22.0 against 24.6). The spread between the strongest and weakest capital region is over 16× (Singapore’s 361.0 against Yangon’s 22.0 Mbps), a gap that a table of national averages conceals.

The trajectory is also diverging rather than converging. Singapore nearly doubled its median provincial fixed speed over the period, from roughly 280 to 550 Mbps, and Thailand grew from roughly 200 to 300. Vietnam accelerated to join Malaysia above 200 Mbps. Laos, Cambodia, Indonesia and Myanmar remained below 60 Mbps for all three years.

The fixed–mobile gap also varies widely. Fixed broadband exceeds mobile by 3.2× in Thailand and 2.1× in Singapore, but the two are near parity in Malaysia and Indonesia (1.1×), and Myanmar is the only country in the set where mobile is *faster* than fixed (0.8×) — a mobile-first market.

3.3 Application tiers diverge

Raising the bar from HD to UHD video separates the region into two groups. Malaysia, Singapore, Thailand and the Philippines score 100% at the 25 Mbps threshold every quarter, as does Vietnam from Q2 2024 (86% in Q1). Cambodia ranges from 67% to 86%, Indonesia improves from 36% to 65%, and Myanmar stays between 39% and 55%. Laos is volatile across the full 50–100% range, which

we attribute to its small measurement volume rather than to genuine quarter-to-quarter swings in capacity.

Cloud gaming separates them further (Figure 2). Singapore’s median clears the 44 Mbps bar in every quarter (100%), and Thailand and Malaysia score 87–100%. The Philippines ranges from 58% to 94%, and Vietnam climbs from under 50% to 94–100% in 2025. At the other end, Cambodia reaches only 38–49%, Laos swings between 29% and 85%, Indonesia peaks at 37% and Myanmar at 31% — meaning their median download speed reaches barely a third of the cloud-gaming bandwidth requirement. Networks whose median clears the voice bar everywhere can still fall far short of what a bandwidth-hungry interactive application needs.

4 ONGOING AND FUTURE WORK

Three directions are in progress. First, we are fitting province-level models of speed against economic and density covariates to separate infrastructure effects from wealth effects. Second, we are extending the NDT7 analysis to peak-hour behavior, where busy-versus-quiet *ratios* remain interpretable even though absolute latency is not, given offshore server placement. Third, we are quantifying the measurement bias introduced by thin-sample provinces, which currently inflate the apparent performance of several emerging markets.

The broader argument we wish to discuss at the poster session is that a speed index answers the wrong question. Basic access across Southeast Asia is effectively universal; what separates these markets is whether their networks survive contact with a demanding application, and that distinction points at different infrastructure investment than a ranking does.

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